Data Profiling Report

1. Introduction

This report presents the data profiling of ten PostgreSQL staging tables. The tables are extracts of SAP finance data and are named BKPF, BSEG, CSKS, CSKT, KNA1, LFA1, SKA1, SKAT, T001 and T002. In total they contain 1,465,129 rows. The purpose of profiling is to understand the structure of each table, how complete it is and how much of it is repeated, before any cleaning or transformation is planned. The report only describes the data. Whether the data is acceptable against business rules is a separate question and belongs to the data quality report.

2. Approach

All ten tables were profiled with the same Jupyter notebook, which uses pandas and SQLAlchemy and runs only SELECT queries, so nothing in the staging tables was changed, filled or deleted. For every table the notebook reads the column names, data types and nullability from information_schema, counts rows and columns, counts missing values and blank strings in each column, counts rows that are exact duplicates of another row (all columns equal), and gives summary statistics for the numeric columns. The date profiling and candidate key sections of the notebook were left empty when it was run, so those two checks are not part of this report. This is explained again in Section 6.

3. Overall Summary

Table 1 gives the main figures for all ten tables. Duplicate rows are counted after leaving out the first copy of each group, and the last column counts every row that belongs to a group of identical rows.

Two points stand out from this summary. First, none of the ten tables has a blank string in any column, so missing information is stored as NULL and not as empty text. Second, almost all of the duplication sits in one table: SKAT contributes 71,924 of the 72,992 duplicate rows, which is 98.5 percent. LFA1 (12.82 percent) and T001 (11.07 percent) have the highest duplicate rates after SKAT, while SKA1 and T002 have no exact duplicates at all. It was also seen in the schema output that every column in every table is nullable, so the staging layer does not enforce any constraints.

Table 1: Summary of the ten staging tables

Table	Description	Rows	Cols	Cols with NULLs	Dup. rows	Dup. %	Rows in groups
BKPF	Document header	150,057	15	0	470	0.31	921
BSEG	Document line items	332,106	21	13	77	0.02	154
CSKS	Cost center master	3,353	9	5	40	1.19	79
CSKT	Cost center texts	26,977	7	2	2	0.01	4
KNA1	Customer master	20,209	12	8	23	0.11	42
LFA1	Vendor master	3,035	12	9	389	12.82	488
SKA1	G/L account master	146,692	11	4	0	0.00	0
SKAT	G/L account texts	782,051	10	4	71,924	9.20	126,461
T001	Company codes	605	9	5	67	11.07	92
T002	Language keys	44	7	0	0	0.00	0
Total		**1,465,129**			**72,992**		

4. Table-wise Observations

4.1 BKPF (accounting document header)

BKPF has 150,057 rows and 15 columns. No column contains missing values or blank strings. There are 470 duplicate rows (0.31 percent) and 921 rows in duplicate groups. In the sample of duplicates that the notebook displays, the repeated rows come from several document types (SA, DZ and RV) and different company codes, so they do not look limited to one source. The only numeric columns are fiscal_year and fiscal_period. Fiscal year runs from 2018 to 2023 with a mean of 2020.11 and a median of 2020, and fiscal period runs from 1 to 12 with a mean of 5.10 and a median of 4. The columns document_date, posting_date and entry_date have the DATE type, while entry_time is stored as text.

4.2 BSEG (accounting document line items)

BSEG has 332,106 rows and 21 columns and is the sparsest of the transactional tables. Thirteen columns have missing values, as listed in Table 2.

Table 2: Missing values in stg_bseg

Column	Missing	% missing
tax_code	332,106	100.00
special_gl_indicator	332,101	100.00
line_item_text	332,054	99.98
cost_center_id	332,026	99.98
clearing_document_number	328,652	98.96
clearing_date	328,652	98.96
customer_id	293,496	88.37
payment_terms	258,029	77.69
payement_baseline_date	257,694	77.59
vendor_id	144,799	43.60
assignment_no	134,520	40.51
amount_document_currency	92	0.03
amount_local_currency	3	0.00

The column tax_code is empty in every row and special_gl_indicator has a value in only 5 rows. Only 3,454 rows (1.04 percent) carry clearing information. customer_id is filled in 38,610 rows and vendor_id in 187,307 rows. There are 77 exact duplicate rows (0.02 percent). The two amount columns have a minimum of 0.03, a median of about 59,700, a mean of about 14.8 million and a maximum of about 98.98 billion, with a standard deviation of about 647 million. The mean is roughly 250 times the median, so the distribution is very heavily skewed and the largest values should be checked with the business. No negative amounts are present, so the sign is probably given by debit_credit_indicator. The column name payement_baseline_date is misspelled, and it is stored as text together with clearing_date.

4.3 CSKS (cost center master)

CSKS has 3,353 rows and 9 columns. The columns valid_from_date, company_code, cost_center_category, responsible_person and currency_code each have exactly one NULL (0.03 percent). Since the counts are the same, they may belong to a single record, but this was not checked. There are 40 duplicate rows (1.19 percent) and 79 rows in duplicate groups. The table has no numeric columns, and both validity dates are of DATE type.

4.4 CSKT (cost center texts)

CSKT has 26,977 rows and 7 columns. cost_center_description is missing in 879 rows (3.26 percent) and cost_center_name in 282 rows (1.05 percent). There are only 2 duplicate rows (0.01 percent), which form 4 rows in duplicate groups. There are no numeric columns.

4.5 KNA1 (customer master)

KNA1 has 20,209 rows and 12 columns, and eight of the columns have missing values (Table 3).

Table 3: Missing values in stg_kna1

Column	Missing	% missing
customer_name_2	19,263	95.32
street_address	5,081	25.14
regional_code	2,977	14.73
postal_code	622	3.08
city	155	0.77
language_code	25	0.12
country_code	2	0.01
customer_name	2	0.01

There are 23 duplicate rows (0.11 percent) and 42 rows in duplicate groups. client_id is stored as BIGINT and takes values from 50 to 800. The median and the upper quartiles are all 800 and the mean is 785.03, so most of the customers belong to client 800. created_date is a TIMESTAMP and has no missing values.

4.6 LFA1 (vendor master)

LFA1 has 3,035 rows and 12 columns, with nine columns that contain missing values (Table 4).

Table 4: Missing values in stg_lfa1

Column	Missing	% missing
vendor_name2	2,573	84.78
region_code	1,367	45.04
street_address	458	15.09
postal_code	420	13.84
city	250	8.24
language_code	43	1.42
vendor_name	5	0.16
country_code	3	0.10
created_date	1	0.03

LFA1 has 389 duplicate rows, which is 12.82 percent of the table and the highest rate among the master tables, and 488 rows belong to duplicate groups. client_id is BIGINT with values from 50 to 800, a mean of 700.76 and a median of 800, so the table mixes more than one client. created_date is a TIMESTAMP.

4.7 SKA1 (G/L account master)

SKA1 has 146,692 rows and 11 columns. Four columns have missing values (Table 5). There are no duplicate rows. The notebook for this table was connected to a database called ska1new, while the other nine used a database called data_profiling.

Table 5: Missing values in stg_ska1

Column	Missing	% missing
financial_statement_type	86,992	59.30
balance_sheet_indicator	59,700	40.70
created_by_user	4,060	2.77
created_date	4,060	2.77

created_by_user and created_date have the same number of NULLs, which suggests that the same rows are affected. There are no numeric columns. created_date and record_timestamp are stored as text and is_deleted is BOOLEAN. Because the table also has operation_flag, is_deleted and record_timestamp, it may hold change records, which would explain the high row count for an account master. This should be confirmed with the source team.

4.8 SKAT (G/L account texts)

SKAT is the largest table, with 782,051 rows and 10 columns. Four columns have missing values (Table 6).

Table 6: Missing values in stg_skat

Column	Missing	% missing
chart_of_accounts	782,051	100.00
search_term	20,709	2.65
gl_account_description	20,239	2.59
gl_account_short_name	16,666	2.13

chart_of_accounts is empty in every row, and pandas reads it as an object column for that reason. The more serious finding is the duplication: 71,924 rows (9.20 percent) repeat an earlier row, and 126,461 rows belong to a duplicate group. This is the main data problem in the whole set of tables. There are no numeric columns. record_timestamp is stored as text and is_deleted is BOOLEAN.

4.9 T001 (company codes)

T001 has 605 rows and 9 columns, and five columns have missing values (Table 7).

Table 7: Missing values in stg_t001

Column	Missing	% missing
country_code	24	3.97
chart_of_accounts	16	2.64
language_code	12	1.98
city	8	1.32
fiscal_year_variant	8	1.32

There are 67 duplicate rows (11.07 percent) and 92 rows in duplicate groups. The duplicates displayed by the notebook are template company codes (ID01, QA01, AE01, HK01, IE01, MY01, NZ01, PH01, SG01 and BE01), each of which appears twice with identical values. There are no numeric columns.

4.10 T002 (language keys)

T002 has 44 rows and 7 columns. It has no missing values, no blank strings and no duplicate rows. The only numeric column is language_hierarchy_level, which is an integer from 0 to 4 with a mean of 1.25, a standard deviation of 1.60 and a median of 0. is_deleted is stored as text here, although SKA1 and SKAT store it as BOOLEAN, and record_timestamp is also text.

5. Issues Seen Across the Tables

Missing data. Two columns are completely empty, BSEG.tax_code and SKAT.chart_of_accounts, and BSEG.special_gl_indicator has only five values. Some optional fields are nearly unused: BSEG.line_item_text and cost_center_id are filled in only about 0.02 percent of rows, KNA1.customer_name_2 in 4.68 percent and LFA1.vendor_name2 in 15.22 percent. Other fields look as if they should always be present but have gaps, namely customer_name (2 NULLs in KNA1), vendor_name (5 in LFA1), country_code (24 NULLs in T001, 3 in LFA1 and 2 in KNA1) and the BSEG amounts (3 and 92 NULLs).

Duplicates. In total 72,992 rows repeat an earlier row. The notebook does not assume a business key, so only rows that are identical in every column were counted. Rows that share a key but differ in other columns were not examined.

Data types and names. client_id is text in BKPF, BSEG, CSKS, CSKT, SKA1, SKAT and T001, but BIGINT in KNA1 and LFA1, which can cause problems when these tables are joined. Dates and times are stored in three different ways: DATE (BKPF, CSKS, CSKT), TIMESTAMP (KNA1, LFA1) and text (BSEG, SKA1, SKAT, T002). is_deleted is BOOLEAN in SKA1 and SKAT but text in T002. Field names are also not uniform: gl_account_no is used in BSEG and SKA1 but gl_account_number in SKAT, regional_code in KNA1 corresponds to region_code in LFA1, and BSEG has the misspelled payement_baseline_date.

6. Limitations

The conclusions above are limited by what the notebooks actually ran. No date checks were done because the DATE_COLUMNS list was empty in all ten notebooks, so the earliest and latest dates and any values that cannot be parsed are unknown, which matters most for the date columns stored as text. No key check was done because CANDIDATE_KEY was also empty, so uniqueness of the business keys is not confirmed. The row counts in staging were not compared with the source systems. Distinct counts, value frequencies, format patterns and placeholder values such as N/A or 0 were not profiled, so a column without NULLs may still contain unusable values. Relationships between tables, for example BSEG to BKPF or SKAT to SKA1, were not examined.

7. Suggested Next Steps

None of the following should be applied directly to the staging tables. Any cleaned data should be created as a separate dataset once the rule behind it has been justified.

1. Confirm the business keys from the SAP data dictionary and run the candidate key check. For example, client with company code, document number and fiscal year for BKPF (plus line item number for BSEG), client with company code for T001, client with customer for KNA1 and client with vendor for LFA1.
2. Find out why SKAT, LFA1 and T001 contain so many identical rows (repeated extracts or change records) before deciding on any rule for removing them.
3. Check the very large BSEG amounts, including the maximum of about 98.98 billion, with the finance team.
4. Ask the source team whether BSEG.tax_code and SKAT.chart_of_accounts are really unused or were lost during extraction.
5. Run the date profiling for all date columns, especially those stored as text.

6. Make client_id, the date columns and is_deleted consistent in the transformation layer, and correct the name payement_baseline_date there as well.
7. Compare the staging row counts with the source systems.

8. Conclusion

The ten tables are structurally readable and contain no blank strings, but three problems repeat across them: many identical rows (mainly in SKAT, LFA1 and T001), columns that are empty or almost empty (mainly in BSEG and SKAT), and data types and column names that differ between tables. T002 showed no issue in any of the checks that were run, and SKA1 has no duplicate rows. The next stage is the data quality report, where confirmed keys and business rules can be used to decide which of these observations are real failures.